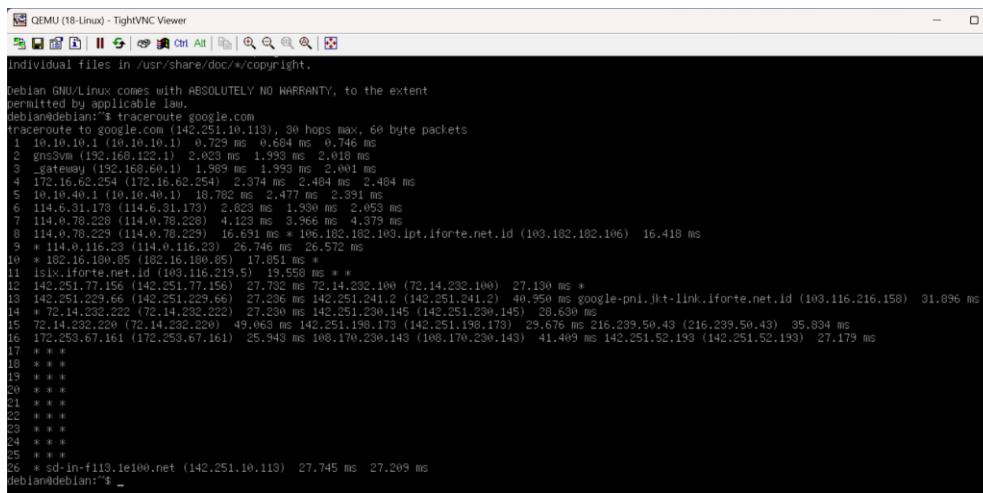


Praktikum Jaringan Komputer
Praktikum 6 Protokol Lapisan Network 2



Disusun Oleh:
Nabila Nur 'Abidah Putri Valiandra
244107060086
D-IV Sistem Informasi Bisnis
2C

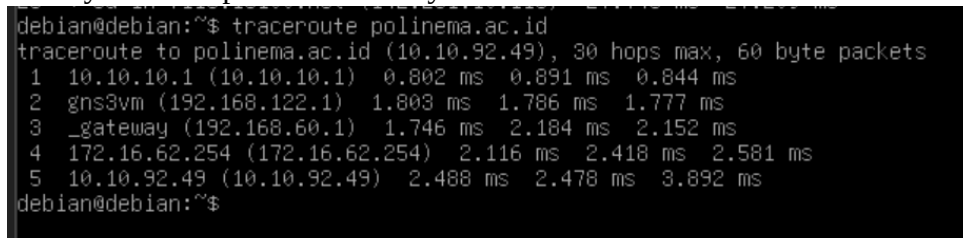
1. Gunakan program traceroute untuk melacak gateway yang dilalui ke host luar jaringan intranet Polinema (facebook, google atau yang lainnya). Amati hasilnya dan simpan screenshotnya.



```
individual files in /usr/share/doc/*/copyright.
debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
debian@debian:~$ traceroute google.com
traceroute to google.com (142.251.10.113), 30 hops max, 60 byte packets
 1 10.10.10.1 (10.10.10.1) 0.723 ms 0.684 ms 0.746 ms
 2 gns3vm (192.168.122.1) 2.023 ms 1.993 ms 2.018 ms
 3 _gateway (192.168.60.1) 1.989 ms 1.993 ms 2.001 ms
 4 172.16.62.254 (172.16.62.254) 2.374 ms 2.484 ms 2.484 ms
 5 10.10.40.1 (10.10.40.1) 18.782 ms 2.477 ms 2.391 ms
 6 114.6.31.173 (114.6.31.173) 2.823 ms 1.930 ms 2.053 ms
 7 114.0.78.228 (114.0.78.228) 4.123 ms 3.966 ms 4.379 ms
 8 114.0.78.229 (114.0.78.229) 16.691 ms * 106.182.182.103.lpt.lforte.net.id (103.182.182.106) 16.418 ms
 9 * 114.0.116.23 (114.0.116.23) 26.746 ms 26.572 ms
10 * 182.16.180.85 (182.16.180.85) 17.851 ms *
11 isix.lforte.net.id (103.116.219.5) 19.558 ms * *
12 142.251.77.156 (142.251.77.156) 27.732 ms 72.14.232.100 (72.14.232.100) 27.130 ms *
13 142.251.229.66 (142.251.229.66) 27.236 ms 142.251.241.2 (142.251.241.2) 40.950 ms google-pnl.jkt-link.lforte.net.id (103.116.216.158) 31.896 ms
14 * 72.14.232.222 (72.14.232.222) 27.230 ms 142.251.230.145 (142.251.230.145) 28.630 ms
15 72.14.232.220 (72.14.232.220) 49.063 ms 142.251.198.173 (142.251.198.173) 29.676 ms 216.239.50.43 (216.239.50.43) 35.834 ms
16 172.253.67.161 (172.253.67.161) 25.943 ms 108.170.230.143 (108.170.230.143) 41.409 ms 142.251.52.193 (142.251.52.193) 27.179 ms
17 * *
18 * *
19 * *
20 * *
21 * *
22 * *
23 * *
24 * *
25 * *
26 * sd-in-f113.1e100.net (142.251.10.113) 27.745 ms 27.209 ms
debian@debian:~$
```

Penjelasan: Traceroute menampilkan jalur (gateway) yang dilewati paket data dari komputer kita menuju ke server tujuan (misalnya google.com). Setiap “hop” adalah router atau gateway yang dilewati.

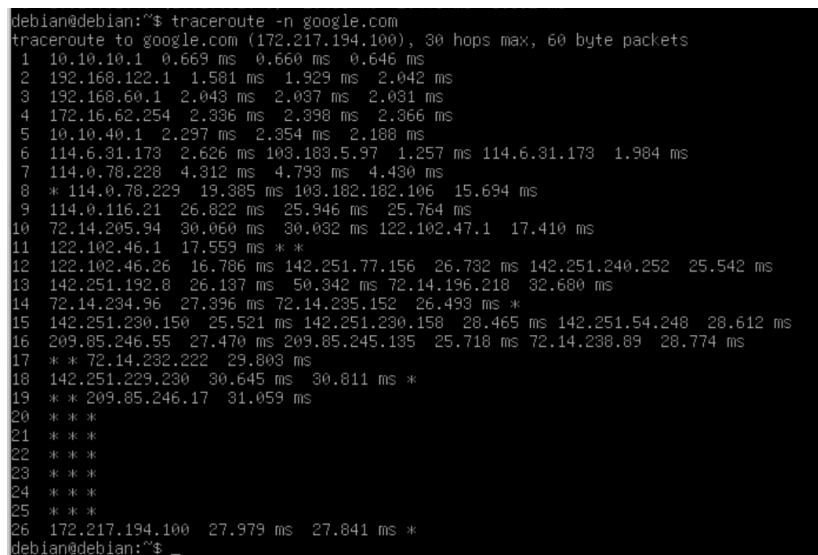
2. Gunakan program traceroute untuk melacak gateway yang dilalui ke polinema.ac.id. Amati hasilnya dan simpan screenshotnya.



```
debian@debian:~$ traceroute polinema.ac.id
traceroute to polinema.ac.id (10.10.92.49), 30 hops max, 60 byte packets
 1 10.10.10.1 (10.10.10.1) 0.802 ms 0.891 ms 0.844 ms
 2 gns3vm (192.168.122.1) 1.803 ms 1.786 ms 1.777 ms
 3 _gateway (192.168.60.1) 1.746 ms 2.184 ms 2.152 ms
 4 172.16.62.254 (172.16.62.254) 2.116 ms 2.418 ms 2.581 ms
 5 10.10.92.49 (10.10.92.49) 2.488 ms 2.478 ms 3.892 ms
debian@debian:~$
```

Jawaban: Traceroute ke domain lokal menunjukkan bahwa lokasi server lebih dekat dan jalur routing lebih pendek dibanding ke host global seperti Google.

3. Jelaskan arti dari hasil traceroute tersebut

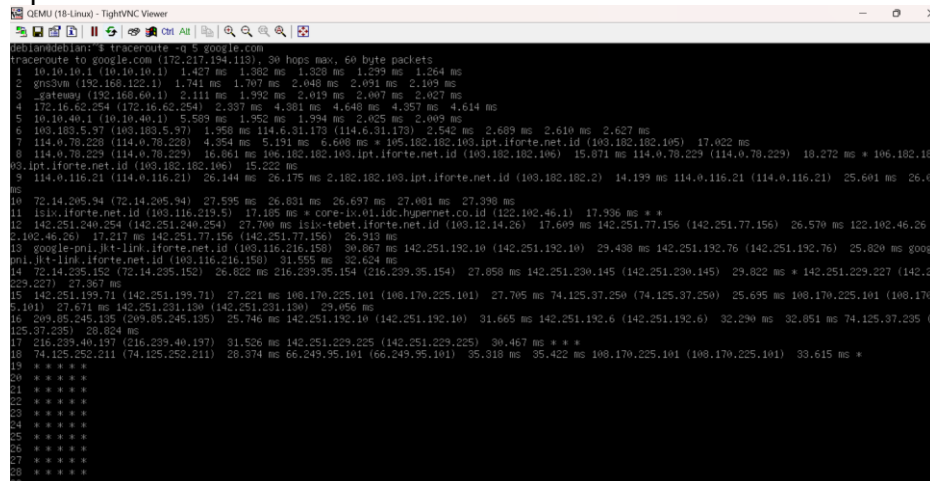


```
debian@debian:~$ traceroute -n google.com
traceroute to google.com (172.217.194.100), 30 hops max, 60 byte packets
 1 10.10.10.1 0.669 ms 0.660 ms 0.646 ms
 2 192.168.122.1 1.581 ms 1.929 ms 2.042 ms
 3 192.168.60.1 2.043 ms 2.037 ms 2.031 ms
 4 172.16.62.254 2.336 ms 2.398 ms 2.366 ms
 5 10.10.40.1 2.297 ms 2.354 ms 2.188 ms
 6 114.6.31.173 2.626 ms 103.183.5.97 1.257 ms 114.6.31.173 1.984 ms
 7 114.0.78.228 4.312 ms 4.793 ms 4.430 ms
 8 * 114.0.78.229 19.385 ms 103.182.182.106 15.694 ms
 9 114.0.116.21 26.822 ms 25.946 ms 25.764 ms
10 72.14.205.94 30.060 ms 30.032 ms 122.102.47.1 17.410 ms
11 122.102.46.1 17.559 ms * *
12 122.102.46.26 16.786 ms 142.251.77.156 26.732 ms 142.251.240.252 25.542 ms
13 142.251.192.8 26.137 ms 50.342 ms 72.14.196.218 32.680 ms
14 72.14.234.96 27.396 ms 72.14.235.152 26.493 ms *
15 142.251.230.150 25.521 ms 142.251.230.158 28.465 ms 142.251.54.248 28.612 ms
16 209.85.246.55 27.470 ms 209.85.245.135 25.718 ms 72.14.238.89 28.774 ms
17 * * 72.14.232.222 29.803 ms
18 142.251.229.230 30.645 ms 30.811 ms *
19 * * 209.85.246.17 31.059 ms
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 172.217.194.100 27.979 ms 27.841 ms *
debian@debian:~$
```

Jawaban: Perintah traceroute -n google.com digunakan untuk melihat jalur yang dilewati paket data dari komputer ke server Google. Hasilnya menunjukkan beberapa hop dari jaringan lokal (10.10.10.x, 192.168.x.x) menuju jaringan ISP hingga ke server Google (172.217.194.100) dengan waktu rata-rata sekitar 27 ms. Tanda * menandakan router yang tidak merespons, tapi koneksi tetap berjalan normal dan stabil.

4. Gunakan program traceroute untuk 4 parameter lainnya ke host tujuan (terserah), amati perbedaannya dan simpan screenshotnya

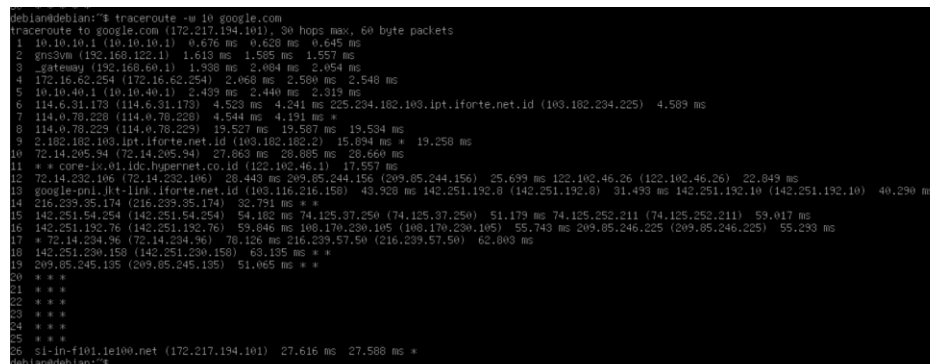
-q



```
sublan@debian:~$ traceroute -q 5 google.com
traceroute to google.com (172.217.194.113), 30 hops max, 60 byte packets
 1 10.10.10.1 (10.10.10.1) 1.427 ms 1.382 ms 1.328 ms 1.299 ms 1.264 ms
 2 gns3vm (192.168.122.1) 1.741 ms 1.707 ms 2.048 ms 2.051 ms 2.105 ms
 3 gateway (192.168.60.1) 2.111 ms 1.992 ms 2.019 ms 2.007 ms 2.027 ms
 4 172.16.62.254 (172.16.62.254) 2.337 ms 4.381 ms 4.640 ms 4.357 ms 4.614 ms
 5 10.10.40.1 (10.10.40.1) 5.589 ms 1.952 ms 1.934 ms 2.025 ms 2.009 ms
 6 103.182.5.97 (103.182.5.97) 1.958 ms 114.6.31.173 (114.6.31.173) 2.542 ms 2.689 ms 2.610 ms 2.627 ms
 7 114.0.78.229 (114.0.78.229) 4.354 ms 5.191 ms 5.608 ms * 105.182.182.103.lpt.iforte.net.id (103.182.182.105) 17.022 ms
 8 114.0.78.229 (114.0.78.229) 16.861 ms 106.182.182.103.lpt.iforte.net.id (103.182.182.106) 15.871 ms 114.0.78.229 (114.0.78.229) 10.272 ms * 106.182.182.103.lpt.iforte.net.id (103.182.182.106) 15.222 ms
 9 114.0.116.21 (114.0.116.21) 26.144 ms 26.175 ms 2.182.182.103.lpt.iforte.net.id (103.182.182.2) 14.199 ms 114.0.116.21 (114.0.116.21) 25.691 ms 26.603 ms
10 72.14.205.94 (72.14.205.94) 27.595 ms 26.831 ms 26.637 ms 27.081 ms 27.398 ms
11 131x.iforte.net.id (103.116.219.5) 17.185 ms * core-ix-01.idc.hypernet.co.id (122.182.46.1) 17.396 ms *
12 142.251.230.154 (142.251.230.154) 27.700 ms 131x-toilet.iforte.net.id (103.12.14.26) 17.609 ms 142.251.77.156 (142.251.77.156) 26.570 ms 122.102.46.26 (122.102.46.26) 17.217 ms 142.251.77.156 (142.251.77.156) 26.913 ms
13 google-pnl-jkt-link.iforte.net.id (103.116.216.158) 30.867 ms 142.251.192.10 (142.251.192.10) 29.438 ms 142.251.192.76 (142.251.192.76) 25.820 ms google-pnl-jkt-link.iforte.net.id (103.116.216.158) 31.555 ms 32.624 ms
14 72.14.239.152 (72.14.239.152) 26.822 ms 216.239.35.154 (216.239.35.154) 27.858 ms 142.251.230.145 (142.251.230.145) 29.822 ms * 142.251.229.227 (142.251.229.227) 27.367 ms
15 142.251.199.71 (142.251.199.71) 27.221 ms 108.170.225.101 (108.170.225.101) 27.705 ms 74.125.37.250 (74.125.37.250) 25.695 ms 108.170.225.101 (108.170.225.101) 27.671 ms 142.251.231.130 (142.251.231.130) 23.456 ms
16 209.85.245.135 (209.85.245.135) 55.746 ms 142.251.192.10 (142.251.192.10) 31.665 ms 142.251.192.6 (142.251.192.6) 32.290 ms 32.851 ms 74.125.37.235 (74.125.37.235) 28.824 ms
17 216.239.40.197 (216.239.40.197) 31.526 ms 142.251.229.225 (142.251.229.225) 30.467 ms * *
18 74.125.252.211 (74.125.252.211) 20.374 ms 66.249.95.101 (66.249.95.101) 35.318 ms 35.422 ms 108.170.225.101 (108.170.225.101) 33.615 ms *
19 * * * * *
20 * * * * *
21 * * * * *
22 * * * * *
23 * * * * *
24 * * * * *
25 * * * * *
26 * * * * *
```

Analisis: Jumlah dari query per hop diubah dari 3 menjadi 5. Artinya setiap hop akan diuji sebanyak 5 kali, maka hasilnya lebih akurat tapi waktu eksekusi akan lebih lama.

-w



```
sublan@debian:~$ traceroute -w 10 google.com
traceroute to google.com (172.217.194.101), 30 hops max, 60 byte packets
 1 10.10.10.1 (10.10.10.1) 0.676 ms 0.620 ms 0.645 ms
 2 gns3vm (192.168.122.1) 1.613 ms 1.585 ms 1.557 ms
 3 gateway (192.168.60.1) 1.938 ms 2.084 ms 2.054 ms
 4 172.16.62.254 (172.16.62.254) 2.068 ms 2.580 ms 2.548 ms
 5 10.10.40.1 (10.10.40.1) 2.439 ms 2.440 ms 2.319 ms
 6 114.6.31.173 (114.6.31.173) 4.523 ms 4.241 ms 225.234.182.103.lpt.iforte.net.id (103.182.234.225) 4.589 ms
 7 114.0.78.229 (114.0.78.229) 4.544 ms 4.191 ms *
 8 114.0.78.229 (114.0.78.229) 19.527 ms 19.587 ms 19.534 ms
 9 2.182.182.103.lpt.iforte.net.id (103.182.182.2) 15.894 ms * 19.258 ms
10 72.14.205.94 (72.14.205.94) 27.863 ms 28.885 ms 28.660 ms
11 * * core-ix-01.idc.hypernet.co.id (122.182.46.1) 17.557 ms
12 72.14.232.106 (72.14.232.106) 28.443 ms 209.85.244.156 (209.85.244.156) 25.689 ms 122.102.46.26 (122.102.46.26) 22.849 ms
13 google-pnl-jkt-link.iforte.net.id (103.116.216.158) 45.328 ms 142.251.192.8 (142.251.192.8) 31.493 ms 142.251.192.10 (142.251.192.10) 40.290 ms
14 216.239.35.174 (216.239.35.174) 32.791 ms *
15 142.251.54.254 (142.251.54.254) 54.182 ms 74.125.37.250 (74.125.37.250) 51.179 ms 74.125.252.211 (74.125.252.211) 59.017 ms
16 142.251.192.76 (142.251.192.76) 59.846 ms 108.170.230.105 (108.170.230.105) 55.743 ms 209.85.246.225 (209.85.246.225) 55.293 ms
17 * 72.14.234.96 (72.14.234.96) 78.126 ms 216.239.57.50 (216.239.57.50) 62.803 ms
18 142.251.230.158 (142.251.230.158) 63.135 ms *
19 209.85.245.135 (209.85.245.135) 51.065 ms *
20 * *
21 * *
22 * *
23 * *
24 * *
25 * *
26 51-in-f100.net (172.217.194.101) 27.616 ms 27.588 ms *
```

Analisis: mengubah waktu tunggu respon dari yang defaultnya 5 detik diubah menjadi 10 detik. Ini digunakan saat jaringan lambat agar traceroute tidak timeout terlalu cepat.

-f

```

debian@debian:~$ traceroute -f 3 google.com
traceroute to google.com (172.217.194.113), 30 hops max, 60 byte packets
 0  _gateway [192.168.60.1] 113.186 ms 113.169 ms 113.316 ms
 1  172.16.62.254 (172.16.62.254) 113.300 ms 113.335 ms 113.320 ms
 2  10.10.40.1 (10.10.40.1) 113.360 ms 113.310 ms 113.326 ms
 3  114.6.31.173 (114.6.31.173) 113.715 ms 114.079 ms 114.042 ms
 4  * * *
 5  * 114.0.78.229 (114.0.78.229) 113.882 ms 106.182.182.103.lpt.iftorte.net.id (103.182.182.106) 84.094 ms
 6  * 114.0.116.21 (114.0.116.21) 89.492 ms 93.279 ms 2.182.182.103.lpt.iftorte.net.id (103.182.182.2) 85.322 ms
 7  72.14.205.94 (72.14.205.94) 95.000 ms 182.16.180.85 (182.16.180.85) 84.125 ms 83.559 ms
 8  * * *
 9  122.182.46.26 (122.182.46.26) 85.955 ms 85.712 ms 142.251.230.253 (142.251.230.252) 98.250 ms
10  142.251.192.10 (142.251.192.10) 95.760 ms google-pm1.jkt1.lind.iftorte.net.id (103.116.216.150) 100.762 ms 142.251.192.12 (142.251.192.12) 98.406 ms
11  * 142.251.230.137 (142.251.230.137) 97.765 ms 142.251.229.225 (142.251.229.225) 95.373 ms
12  142.251.230.158 (142.251.230.158) 99.940 ms 209.85.242.109 (209.85.242.109) 89.565 ms 100.718 ms
13  209.85.246.225 (209.85.246.225) 100.566 ms 74.125.37.249 (74.125.37.249) 100.819 ms 209.85.246.55 (209.85.246.55) 100.624 ms
14  * * 72.14.234.96 (72.14.234.96) 121.990 ms
15  * * *
16  * * 66.249.94.185 (66.249.94.185) 114.948 ms
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  sf-in-f113.1e100.net (172.217.194.113) 54.128 ms 54.395 ms 54.697 ms
debian@debian:~$

```

Analisis: memulai traceroute dari TTL 3 bukan dari 1 yang artinya hop 1 dan hop 2 dilewati tanpa diuji dan cocok digunakan jika kita hanya ingin menganalisis jalur tengah/akhir jaringan.

5. Gunakan sistem operasi windows dan program tracert untuk melacak gateway yang dilalui ke host di luar jaringan intranet polinema (facebook, google, atau yang lainnya) dan ke polinema.ac.id. Amati hasilnya dan simpan screenshotsnya.

```

C:\Users\windows>tracert google.com

Tracing route to google.com [74.125.24.138]
over a maximum of 30 hops:
 0  13 ms 26 ms 32 ms 10.10.10.1
 1  19 ms 8 ms 11 ms gns3vm [192.168.122.1]
 2  31 ms 23 ms 38 ms _gateway [192.168.60.1]
 3  53 ms 40 ms 36 ms 172.16.62.254
 4  85 ms 121 ms * 10.10.40.1
 5  1 ms 2 ms 4 ms 114.6.31.173
 6  4 ms 3 ms 3 ms 114.0.78.228
 7  16 ms 16 ms 16 ms 114.0.78.229
 8  26 ms 32 ms 30 ms 114.0.116.23
 9  39 ms 24 ms 24 ms 74.125.48.68
10  27 ms 27 ms 27 ms 172.253.70.41
11  30 ms 32 ms 25 ms 142.251.230.50
12  36 ms 27 ms 27 ms 142.251.230.147
13  32 ms 29 ms 29 ms 142.251.230.156
14  28 ms 32 ms 31 ms 66.249.94.59
15  * * * Request timed out.
16  * * * Request timed out.
17  * * * Request timed out.
18  * * * Request timed out.
19  * * * Request timed out.
20  * * * Request timed out.
21  * * * Request timed out.
22  * * * Request timed out.
23  * * * Request timed out.
24  * * * Request timed out.
25  69 ms 68 ms 96 ms sf-in-f113.1e100.net [74.125.24.138]

Trace complete.

C:\Users\windows>

```

```

C:\Users\windows>tracert polinema.ac.id

Tracing route to polinema.ac.id [10.10.92.49]
over a maximum of 30 hops:
 0  <1 ms <1 ms <1 ms 10.10.10.1
 1  1 ms 1 ms 1 ms gns3vm [192.168.122.1]
 2  1 ms 1 ms 1 ms _gateway [192.168.60.1]
 3  3 ms 1 ms 1 ms 172.16.62.254
 4  2 ms 1 ms 2 ms 10.10.92.49

Trace complete.

C:\Users\windows>

```

Analisis: Walaupun sintaks berbeda namun fungsi tracert sama dengan traceroute yaitu menampilkan rute dan waktu tempuh paket.

6. Gunakan 3 parameter yang dijelaskan di atas untuk program tracert dan amati serta simpan screenshotsnya.

```

C:\Users\windows>tracert -d google.com

Tracing route to google.com [172.217.194.101]
over a maximum of 30 hops:

  1  <1 ms    <1 ms    <1 ms    10.10.10.1
  2   1 ms    <1 ms    <1 ms    192.168.122.1
  3   1 ms     1 ms     1 ms    192.168.60.1
  4   1 ms     1 ms     1 ms    172.16.62.254
  5   1 ms     1 ms     1 ms    10.10.40.1
  6   2 ms     1 ms     1 ms    114.6.31.173
  7   4 ms     3 ms     3 ms    114.0.78.228
  8  16 ms    16 ms    16 ms    114.0.78.229
  9  25 ms    25 ms    25 ms    114.0.116.21
 10 27 ms    27 ms    27 ms    72.14.205.94
 11 27 ms    24 ms    24 ms    209.85.248.209
 12 26 ms    26 ms    25 ms    142.251.192.76
 13 29 ms    29 ms    28 ms    66.249.95.248
 14 27 ms    27 ms    27 ms    108.170.225.101
 15 27 ms    27 ms    28 ms    72.14.238.89
 16 *        *        *        Request timed out.
 17 *        *        *        Request timed out.
 18 *        *        *        Request timed out.
 19 *        *        *        Request timed out.
 20 *        *        *        Request timed out.
 21 *        *        *        Request timed out.
 22 ^C

C:\Users\windows>

```

Analisis: hasil akan menunjukkan bahwa paket data menuju server Google berhasil mencapai hop ke-15 dengan waktu respon maksimal 28 ms sebelum mengalami request timed out mulai hop ke-16. Jalur awal melewati jaringan lokal dan gateway internal, kemudian menuju jaringan publik milik ISP hingga ke server Google. Request timed out di akhir kemungkinan disebabkan oleh sistem keamanan Google yang menolak balasan ICMP. Secara keseluruhan, koneksi internet tetap stabil dan jalur komunikasi berjalan baik tanpa indikasi gangguan.

```

C:\Users\windows>tracert -h 10 google.com

Tracing route to google.com [172.217.194.138]
over a maximum of 10 hops:

  1  <1 ms    <1 ms    <1 ms    10.10.10.1
  2   1 ms    <1 ms    <1 ms    gns3vm [192.168.122.1]
  3   1 ms     1 ms     1 ms    _gateway [192.168.60.1]
  4   1 ms     1 ms     1 ms    172.16.62.254
  5   1 ms     1 ms     1 ms    10.10.40.1
  6   2 ms     1 ms     1 ms    114.6.31.173
  7   4 ms     4 ms     3 ms    114.0.78.228
  8  16 ms    16 ms    16 ms    114.0.78.229
  9  25 ms    25 ms    25 ms    114.0.116.21
 10 27 ms    26 ms    27 ms    72.14.205.94

Trace complete.

C:\Users\windows>

```

Analisis: hasilnya akan menunjukkan paket data berhasil mencapai server Google melalui 10 hop dengan waktu respon maksimal 27 ms. Jalur awal melewati jaringan lokal dan virtual (IP privat), lalu dilanjutkan ke jaringan publik milik ISP hingga akhirnya ke server Google. Tidak ada keterlambatan atau kegagalan transmisi, menandakan koneksi internet stabil dan jalur komunikasi berjalan dengan baik.

```

C:\Users\windows>tracert -w 1000 google.com

Tracing route to google.com [172.217.194.113]
over a maximum of 30 hops:

  1    1 ms    1 ms    1 ms  10.10.10.1
  2    3 ms    1 ms    2 ms  gns3vm [192.168.122.1]
  3   20 ms   55 ms   36 ms  _gateway [192.168.60.1]
  4   25 ms   18 ms   22 ms  172.16.62.254
  5    7 ms   46 ms   46 ms  10.10.40.1
  6    8 ms    6 ms    9 ms  114.6.31.173
  7    8 ms   21 ms    6 ms  114.0.78.228
  8   26 ms   53 ms   80 ms  114.0.78.229
  9   66 ms   83 ms   73 ms  114.0.116.21
 10   58 ms   69 ms   65 ms  72.14.205.94
 11   61 ms   63 ms   77 ms  172.253.77.225
 12   92 ms   77 ms  101 ms  142.251.192.12
 13   56 ms   49 ms   53 ms  209.85.255.43
 14   29 ms   27 ms   40 ms  216.239.62.42
 15   65 ms   60 ms   51 ms  74.125.37.249
 16    *      *      *    Request timed out.
 17    *      *      *    Request timed out.
 18    *      *      *    Request timed out.
 19    *      *      *    Request timed out.
 20    *      *      *    Request timed out.
 21    *      *      *    Request timed out.
 22    *      *      *    Request timed out.
 23    *      *      *    Request timed out.
 24    *      *      *    Request timed out.
 25    *      *      *    Request timed out.
 26    *      *      *    Request timed out.
 27   58 ms   27 ms   27 ms  si-in-f113.1e100.net [172.217.194.113]

Trace complete.

C:\Users\windows>

```

Analisis: Ini digunakan untuk mengatur waktu tunggu maksimum untuk tiap respon dalam milidetik. Berguna juga saat jaringan lambat atau ada delay besar.